

Inventory Analytics for EV Spare Parts

Project Assignment

1. Business Story

A regional service company provides maintenance and repair services for electric vehicles. The company operates a spare-parts warehouse that supplies several EV service centers. The warehouse stores parts such as battery modules, charging-port components, brake sensors, thermal-management components, inverters, electric motors, cables, connectors, and electronic control units.

Demand for EV spare parts is uncertain. Some parts are requested frequently because they are used in regular maintenance. Other parts are expensive and rarely demanded, but they are critical when a vehicle is waiting for repair. This makes inventory planning difficult. Keeping too many parts in stock creates high inventory holding costs, especially for expensive battery-related and electronic components. Keeping too few parts creates stockouts, delayed repairs, emergency orders, lost service revenue, and customer dissatisfaction.

The warehouse manager wants to improve spare-parts planning using historical demand and inventory data. The objective is not only to forecast demand accurately, but also to understand the operational cost of forecasting errors. A forecast that is too high may lead to excessive inventory and holding cost. A forecast that is too low may lead to understocking and stockout cost.

In this project, you will act as a data analyst for the EV spare-parts warehouse. You will analyze historical spare-parts demand, inventory levels, replenishment patterns, holding costs, stockout costs, and demand forecasts. You will compare forecasting methods using both error-based evaluation and cost-based evaluation.

Central question: Can a forecasting method with low statistical error still create high inventory cost?

2. Data Description

You are given three CSV files. The data are weekly. The variable `week_start` identifies the beginning of each weekly period.

2.1 parts.csv

This file contains information about each EV spare part.

Column	Description
part_id	Unique spare-part identifier
part_name	Name of the spare part
category	Spare-part category
unit_cost	Purchasing cost per unit
unit_price	Selling price or internal transfer price per unit
supplier	Supplier name
lead_time_days	Average supplier lead time in days
demand_trend	Synthetic trend label: Increasing, Decreasing, Stable, or Seasonal
holding_cost_per_unit_week	Weekly holding cost for keeping one unit in inventory
stockout_cost_per_unit	Cost of one unit of unmet demand

Possible categories include Battery System, Charging System, Brake System, Thermal Management, Power Electronics, Electric Motor, Sensors, Cables and Connectors, and Control Units.

2.2 demand.csv

This file contains weekly spare-part demand from EV service centers.

Column	Description
week_start	Beginning of the weekly period
part_id	Spare-part identifier
units_demanded	Number of units requested by service centers during the week

service_center_region	Region of the requesting service center
vehicle_type	EV model group
service_campaign	Whether a special maintenance or recall campaign was active

2.3 inventory.csv

This file contains observed weekly inventory information before demand is fulfilled. You can merge this file with demand.csv to analyze inventory consequences.

Column	Description
week_start	Beginning of the weekly period
part_id	Spare-part identifier
stock_on_hand	Inventory available at the beginning of the week
replenishment	Units received from suppliers during the week

3. Suggested Inventory Calculations

After merging demand.csv and inventory.csv using week_start and part_id, you may compute the following variables to analyze inventory performance.

```

available_inventory = stock_on_hand + replenishment
units_fulfilled = min(available_inventory, units_demanded)
unmet_demand = max(units_demanded - available_inventory, 0)
ending_stock = max(available_inventory - units_demanded, 0)
stockout = 1 if unmet_demand > 0, otherwise 0

```

4. Suggested Analyses

Forecasting methods can be examined through both error-based and cost-based evaluation. Since spare parts are discrete items, forecasts or order quantities should be converted into integer units before being used for inventory decisions.

4.1 Error-Based Forecast Evaluation

Error-based evaluation measures statistical forecast accuracy. You may compute metrics such as MSE, MAE, and MAPE. For MAPE, weeks with zero demand should be excluded to avoid division by zero.

4.2 Cost-Based Forecast Evaluation

Cost-based evaluation measures the business consequence of forecast errors. Overage means the forecast is higher than actual demand, which may create holding cost. Underage means the forecast is lower than actual demand, which may create lost sales, emergency orders, or stockout cost. You should state your assumptions clearly when translating forecast errors into costs.

A useful managerial comparison is whether the most precise forecast is also the most economic forecast.

5. Final Deliverables

5.1 Python Code as a Notebook

Submit one completed notebook: ev_spare_parts_inventory_forecasting.ipynb.

The notebook should include readable code, clear comments, and short explanations of the main steps.

5.2 Suggested (Not Necessary) Ideas/Insights for Presentation and Report

Suggested analyses:

- main demand patterns by part, category, region
- parts with frequent stockouts, large unmet demand, or systematically high ending stock
- high-cost parts where overstocking is especially expensive
- high-criticality parts where understocking is especially costly
- comparison between error-based and cost-based forecast rankings

- operational interpretation of overage and underage errors
- managerial actions for replenishment, monitoring, and inventory prioritization

Suggested calculations:

- demand summaries, revenue summaries, and category-level contribution
- inventory KPIs such as stockout rate, fill rate, unmet demand, and ending stock
- holding cost, stockout cost, and total inventory-related cost
- forecast accuracy metrics such as MSE, MAE, and MAPE
- overage cost, underage cost, and total forecast-related cost
- safety stock, reorder point, and recommended order quantity

5.3 Reorder Recommendation File

Submit one CSV file: reorder_recommendations.csv.

The file should include at least the following columns:

Column
part_id
part_name
category
current_stock
forecasted_weekly_demand
lead_time_days
safety_stock
reorder_point
recommended_order_quantity
recommendation

Recommendations should distinguish between parts requiring immediate replenishment, parts with excessive inventory, parts with high stockout risk, and expensive low-demand parts that should not be overstocked.